AgentForge — AI Cost Analysis

LLM Spend Tracking · Token Consumption · Production Projections · March 2026

Executive Summary

AgentForge uses a **dual-model strategy** to minimize cost while maintaining quality: **Claude Sonnet 4.6** handles initial reasoning and tool selection, then **Claude Haiku 4.5** summarizes tool results. This approach reduces per-query cost by **~44%** compared to Sonnet-only, yielding an average cost of **$0.013 per query**.

> **Total development & testing spend: $1.42 · Average cost per query: $0.013 · Projected cost at 1,000 users: $118/month**

1. Development & Testing Costs

All costs below are actual figures tracked during development via Langfuse observability and the built-in AgentMetrics cost calculator.

1.1 LLM API Costs

Two Claude models are used with Anthropic's pay-per-token pricing. The cost formula implemented in `graph.py` is:

```
cost = (sonnet_input × $3.00 + sonnet_output × $15.00 + haiku_input × $0.80 +
haiku_output × $4.00) / 1,000,000
```

Model	Role	Input Price	Output Price	Avg Cost / Query
Claude Sonnet 4.6	Reasoning & tool selection (iter 0)	$3.00 / 1M tokens	$15.00 / 1M tokens	~$0.009
Claude Haiku 4.5	Summarization (iter 1+)	$0.80 / 1M tokens	$4.00 / 1M tokens	~$0.004

Cost Category	Amount	Notes
Evaluation suite (58 tests)	$0.77	Full suite run on 2026-02-28; 100% pass rate
Ad-hoc dev/test queries (~50)	~$0.65	Manual testing during development
Total LLM API spend	$1.42	All development and testing combined

Cost by Test Category

Category	Tests	Total Cost	Avg Cost	Avg Duration	Description
Happy Path	26	$0.350	$0.0134	4.00s	Standard portfolio, performance, tax, dividend queries
Edge Cases	11	$0.129	$0.0117	3.66s	Invalid symbols, empty input, extreme ranges
Adversarial	10	$0.126	$0.0126	6.14s	Prompt injection, impersonation — all refused
Multi-Step	11	$0.164	$0.0149	6.05s	2–3 tools combined per query
TOTAL	58	$0.769	$0.0132	4.69s	

Dual-Model Savings

If all queries used Sonnet exclusively, the average cost would be **$0.024/query** vs the actual **$0.013/query** with the dual-model approach — a **44% reduction**. For the 58-test eval suite, this saved $0.60 ($1.37 Sonnet-only vs $0.77 actual).

1.2 Token Consumption

Token counts are estimated from the measured costs using the known per-token pricing for each model. Each query typically involves one Sonnet call (reasoning) and one Haiku call (summarization).

Metric	Per Query (avg)	Eval Suite (58 tests)	Est. Dev Total (~108 queries)
Input tokens	~4,200	~243,600	~453,600
Output tokens	~740	~42,900	~79,900
Total tokens	~4,940	~286,500	~533,500
Sonnet tokens (input + output)	~1,800	~104,400	~194,400
Haiku tokens (input + output)	~2,900	~168,200	~313,200

Input/output ratio: ~5.7:1 — input tokens dominate because each query includes the system prompt (~800 tokens), user message, tool definitions, and tool result payloads. Output is concise due to temperature=0 and max_tokens=4,096.

1.3 Number of API Calls

Each user query triggers 2–3 LLM API calls on average: one Sonnet call for reasoning, one or more Haiku calls for summarization, plus tool execution calls to external APIs (Ghostfolio, yfinance). The agent supports up to 10 iterations but typically converges in 1–2.

Call Type	Per Query (avg)	Eval Suite (58)	Est. Dev Total (~108)
Sonnet API calls	1	58	~108
Haiku API calls	~1	~50	~95
Total LLM API calls	~2	~108	~203
Ghostfolio API calls (tools)	~1.5	~87	~162
yfinance API calls (market data)	~0.5	~29	~54
Total external API calls	~2	~116	~216

1.4 Observability Tool Costs

AgentForge uses **Langfuse** (cloud-hosted) for tracing, scoring, and feedback collection. Each agent run logs one trace with generation details, token counts, cost, confidence score, and tool metadata.

Service	Tier	Monthly Cost	Included	Dev Usage
Langfuse	Free (Hobby)	$0	50K observations/month	~400 observations
Ghostfolio	Self-hosted	$0	Unlimited API calls	~160 calls

yfinance	Free tier	$0	Rate-limited	~55 calls
Vercel (widget)	Free (Hobby)	$0	100 GB bandwidth	< 1 GB
Railway (backend)	Starter	~$5	500 hrs / $5 credit	Dev only

Total observability cost during development: $0 (within Langfuse free tier)

Development Cost Summary

Item	Cost	% of Total
LLM API (Anthropic)	$1.42	100%
Observability (Langfuse)	$0.00	0%
External APIs (Ghostfolio, yfinance)	$0.00	0%
Infrastructure (Vercel, Railway)	~$5.00	Hosting only
Total Development Spend	$1.42	LLM costs only

2. Production Cost Projections

The following projections estimate monthly costs at four user scales. All figures are based on measured per-query costs from the evaluation suite and realistic usage assumptions.

Assumptions

Parameter	Value	Rationale
Queries per user per day	3	Financial tools used a few times daily for checks and analysis
Days active per month	22	Weekday usage pattern (trading days)
Queries per user per month	66	3 queries/day × 22 days
Avg input tokens per query	4,200	System prompt (~800) + user message + tool defs + tool results
Avg output tokens per query	740	Structured response with tables, disclaimers
Avg total tokens per query	4,940	Input + output combined
Tool calls per query	1.5 avg	Most queries use 1–2 tools; multi-step uses 2–3
Verification overhead	~0%	Rule-based (no LLM call); adds ~0.1s latency, zero token cost
Sonnet/Haiku split	1:1 calls	First iteration uses Sonnet; subsequent use Haiku
Cost per query	$0.013	Measured average from 58-test evaluation suite

Monthly Cost by User Scale

	100 Users	1,000 Users	10,000 Users	100,000 Users
Queries / month	6,600	66,000	660,000	6,600,000
Total tokens / month	32.6M	326M	3.26B	32.6B
Input tokens	27.7M	277M	2.77B	27.7B

Output tokens	4.9M	49M	490M	4.9B
LLM API calls	~13,200	~132,000	~1,320,000	~13,200,000
LLM cost (Anthropic)	$86	$858	$8,580	$85,800
Langfuse	$0 (free)	$59 (Pro)	$59 (Pro)	$199 (Team)
Infrastructure	$5	$20	$100	$500
Total monthly cost	$91	$937	$8,739	$86,499
Cost per user / month	$0.91	$0.94	$0.87	$0.86

Key insight: LLM API costs represent **95–99%** of total spend at every scale. Infrastructure and observability are negligible. The cost-per-user remains nearly flat (~$0.87–$0.94/month) because there are no significant per-user fixed costs.

LLM Cost Breakdown by Model (at 1,000 users)

Model	Calls / month	Input Tokens	Output Tokens	Monthly Cost	% of LLM Spend
Sonnet 4.6	66,000	99M	19.8M	$594	69%
Haiku 4.5	66,000	165M	26.4M	$238	28%
Verification	66,000	0	0	$0	0%
Total	132,000	264M	46.2M	$858*	100%*

***Note:** Sonnet accounts for 69% of LLM cost despite handling only 50% of calls, because its per-token pricing is 3.75× higher than Haiku. If all calls used Sonnet, the monthly cost at 1,000 users would be **~$1,560** — the dual-model strategy saves **$702/month (45%)**.*

Cost Optimization Levers

- **Prompt caching:** Anthropic's prompt caching can reduce input costs by up to 90% for repeated system prompts. At 1,000 users this could save ~$400/month.
- **Batch API:** For non-real-time queries (e.g., scheduled reports), Anthropic's batch API offers 50% discount. Applicable to ~20% of queries.
- **Shift more to Haiku:** Use Haiku for simple lookups (price checks, calendar) that don't require Sonnet-level reasoning. Could shift ratio to 40/60, saving ~15%.
- **Response caching:** Cache identical queries (e.g., "show my portfolio") for 5–15 minutes. Estimated 10–20% cache hit rate at scale.
- **Token reduction:** Compress tool result payloads before feeding to LLM. Could reduce input tokens by ~20%.

Optimized Cost Projection (at 1,000 users)

Scenario	Monthly LLM Cost	Savings vs Baseline	Notes
Baseline (current)	$858	—	Dual-model, no caching
+ Prompt caching	$510	41%	90% reduction on cached system

			prompt tokens
+ Batch API (20% of queries)	$467	46%	50% discount on batch-eligible queries
+ Response caching (15% hit)	$397	54%	Eliminates LLM calls for cached responses
+ Token compression (20%)	$340	60%	Smaller tool result payloads
Sonnet-only (no optimization)	$1,560	−82%	For comparison: single-model baseline

Per-Query Budget Controls

AgentForge enforces hard limits on every query to prevent cost overruns:

Guard	Limit	Current Setting	Configurable Via
Max cost per query	$0.10	$0.10	AGENT_MAX_COST_USD env var
Max iterations	10 LLM–tool loops	10	AGENT_MAX_ITERATIONS env var
Timeout	30 seconds	30s	AGENT_TIMEOUT_SECONDS env var
Max output tokens	4,096 per call	4,096	Agent config
Temperature	0 (deterministic)	0	Agent config

Worst-case single query: $0.10 (hard cap) · Typical query: $0.013 · Most expensive eval query: $0.017

AgentForge · Cost data from evaluation suite run 2026-02-28 · Pricing: Anthropic API (claude-sonnet-4-6, claude-haiku-4-5) · All projections assume current published token pricing